UNNS-U3 v001: Universal Sequence Embeddings and Structural Preservation in the Reversible SELECT-II Arithmetic Graph

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Abstract

We ask which mathematical structures the reversible SELECT-II graph G (vertices $\mathbb{N}_{>0}$, edges $\{kN, k(N+1)^2\}$) carries *intrinsically*, which it preserves through *faithful coding*, and which require *additional generative information*. Results: (i) marked path histories represent every sequence over $\mathbb{N}_{>0}$ exactly (Proved), while unmarked walks are ambiguous, with ambiguity count exactly 2^{L-1} in the extremal case (Verified-bounded, $L \leq 5$; proof sketch given); (ii) discrete suspension over a subshift conjugates the first-return map to the shift, with entropy given measure-wise by Abramov's formula (Proved; numerically corroborated on 56 embeddings); (iii) G is triangle-free and has odd girth exactly 5 (Proved + certificate); (iv) the graph-intrinsic admissibility shifts $X_r(F)$ cannot encode the golden-mean shift (Proved), whereas full shifts are realizable (Proved). Nothing here modifies the frozen U0–U2, TI, TI-DYN or RN releases.

1 Baseline, terminology and evidence labels

Forward move $x \mapsto x(N+1)^2/N$ requires $N \mid x$; inverse $x \mapsto xN/(N+1)^2$ requires $(N+1)^2 \mid x$. Terminology (fixed throughout): **Representation** (a coding with a decoder), **finite-control generation with declared knowledge** (the controller's information is charged), **intrinsic generation** (only graph structure used). Evidence labels: Proved, Verified-bounded, Registered observation, Exploratory, Prior art. Every admissibility result is classed *intrinsic-to-graph*, *uniform-grammar-extension* or *imported-from-source*. Routing uses the lexicographically first shortest path $P(a, b)$; $P(a, a) = (a, 4a, a)$. Provenance: the SHA-256 digests of the ten prior release archives are recorded in `data/provenance_releases.txt`; only the RN and TI archives were re-hashed in the build environment, the rest are copied from the committed folder (not re-run). Registration (config + code hashes) is in `data/u3_registration.json`.

2 U3-1: marked information

Definition 1. For a sequence s the marked history $\iota(s)$ concatenates $P(s_n, s_{n+1})$ and marks the positions of the s_n.

Theorem 1 (U3.1a, Proved)**.** *$D(\iota(s)) = s$ for every finite or infinite $s \in \mathbb{N}^*_{>0}$; hence G represents every positive-integer sequence. (Marks are read off; G is connected, so every $P(a, b)$ exists.)*

Proposition 2 (U3.1b, unmarked ambiguity)**.** *Let $M(w)$ be the number of marked sequences with routing-compatible markings of the unmarked walk w. In the extremal case (concatenation of lex-first geodesics, whose subpaths are lex-first geodesics) $M(w) = 2^{L-1}$ for L segments. Verified for*

$L \leq 5$ *(exhaustive extremal search) and on all* 4^4 *words of a sample; hence* $\lceil \log_2 M \rceil = L - 1$ *bits of marks are needed, unbounded in* L*. Example* $[1, 4, 16]$*:* $M = 2$*.*

Routing was validated against brute-force BFS for all pairs with $d \leq 5$, 500 decode trials, zero failures.

3 U3-2: discrete suspension

For a subshift (X, σ) with embedding e of the alphabet, roof $r(s) = d_G(e_{s_0}, e_{s_1})$ (2 for repeats). The Kakutani tower has first-return map conjugate to σ (checked on 200 random golden-mean sequences). For ergodic μ, $h_{\mu^r} = h_\mu / \bar{r}$ (Abramov); topological entropy is the supremum over measures, hence $\geq$ the lift value of the Parry measure. *Equality holds iff the roof is constant*: observed exactly for the six embeddings with $d = 2$ (e.g. $(3, 4), (5, 8), (6, 8)$), strict otherwise. **Golden mean** ($\log \varphi$ over base): renewal equation $1 - w - w^d = 0$, $w = \lambda^{-2}$. Over embeddings $\{e_0, e_1\} \subset \{1..8\}$ (56 ordered pairs): period 2 in all cases; expanded-graph and renewal entropies agree to 1.5×10^{-15}; Parry-lift entropy equals the Abramov value to 1.7×10^{-16}; five distinct d values / entropies. E.g. $(1, 2)$: $d = 5$, $h_{top} = 0.14060$, $h_{lift} = 0.13154$, $\bar{r} = 3.6584$. **Embedding dependence of mixing** (three-symbol full shift, 84 triples): period 2 for 53, period 1 for 31; $\{1, 2, 3\}$ has period 2, $\{6, 96, 784\}$ period 1. Hence mixing of the suspension is *not* a property of the base system alone.

4 U3-3: intrinsic admissibility (decisive test)

$X_r(F) = \{s : d_G(s_n, s_{n+1}) \leq r\}$ on a finite vertex window F.

Theorem 3 (Obstruction, Proved)**.** *The pair relation is reflexive and symmetric; therefore every constant sequence is admissible and fixed points of* $X_r(F)$ *number* $|F|$*. The golden-mean shift has one fixed point, so it is not conjugate to any* $X_r(F)$*; no letter-to-letter intrinsic rule encodes it. (Verified: no violations in 60 random windows.)*

Theorem 4 (Realizability, Proved)**.** *Full shifts are realized by cliques of* $G^{(r)}$ *(balls of radius* $\lfloor r/2 \rfloor$*).*

Theorem 5 (Triangle-free, Proved)**.** G *contains no triangle; hence at* $r = 1$ *no graph with a triangle is realizable. Proof: an edge* $\{kN, k(N+1)^2\}$*; a triangle forces integers* a, b, c, t *with* $a + b = (t - 1)(1 - ct) \leq 0 < 2 \leq a + b$*, contradiction (identity checked symbolically; sign check* $t, c < 400$*; no triangle among* $x \leq 3000$*).*

Proposition 6 (Odd girth, Proved by certificate)**.** *The cycle* $6, 24, 96, 784, 27$ *has all five edges in* G*, so* G *is non-bipartite with odd girth* 5*.*

Realizability of the 30 connected graphs on ≤ 5 vertices (Registered observation, bounded search): $r = 1$: 19 excluded by the triangle theorem, 10 realized, $K_{2,3}$ *not found* in the universe $V \leq 10^6$, first vertex ≤ 300; $r = 2$ and $r = 3$: 30/30 realized, all embeddings re-verified by exact distance. "Not found" is not an impossibility claim.

5 U3-4/U3-5: compiled admissibility and cost ledger

An explicit forbidden list (golden mean: 2 bits; no-11-22 over 3 symbols: 8; no-111: 3) can be compiled into a marked-history observer; this information is *introduced*, not intrinsic, and is accounted

separately from route cost C_{route} and certification cost C_{cert}. Embedding (8 bits for a 16-wide window) and radius (3 bits) add further declared knowledge. Alphabet-embedding dependence of structure is shown in §3. All neighbourhoods used were certified under the RN-0 policy (primality proofs only below 3.317×10^{24}).

6 Answer to the programme question

Structure	Status	Class
Any sequence as ordered waypoints	faithful coding with marks (U3.1a)	representation
Unmarked walk $\to$ sequence	ambiguous, 2^{L-1} extremal	needs $L-1$ bits
Dynamics (shift) on the walk	suspension, return map conjugate	coding
Golden-mean admissibility	not intrinsic (obstruction)	needs declared knowledge
Full-shift admissibility	intrinsic via cliques	intrinsic
Local structure (triangle-free, odd girth 5)	intrinsic	graph

7 Limitations and amendments

FIELD-I/GIG preservation (G8) is *not attempted*: exact definitions were not verified. Amendments A1 (Parry-matrix correction), A2 (bounded realizability search), A3 (topological entropy versus Abramov's measure-wise value) are disclosed in `u3_config.json`. Entropy numerics are corroboration, not proof. The 2^{L-1} statement is proved only as a sketch and verified for $L \leq 5$. Searches are bounded.